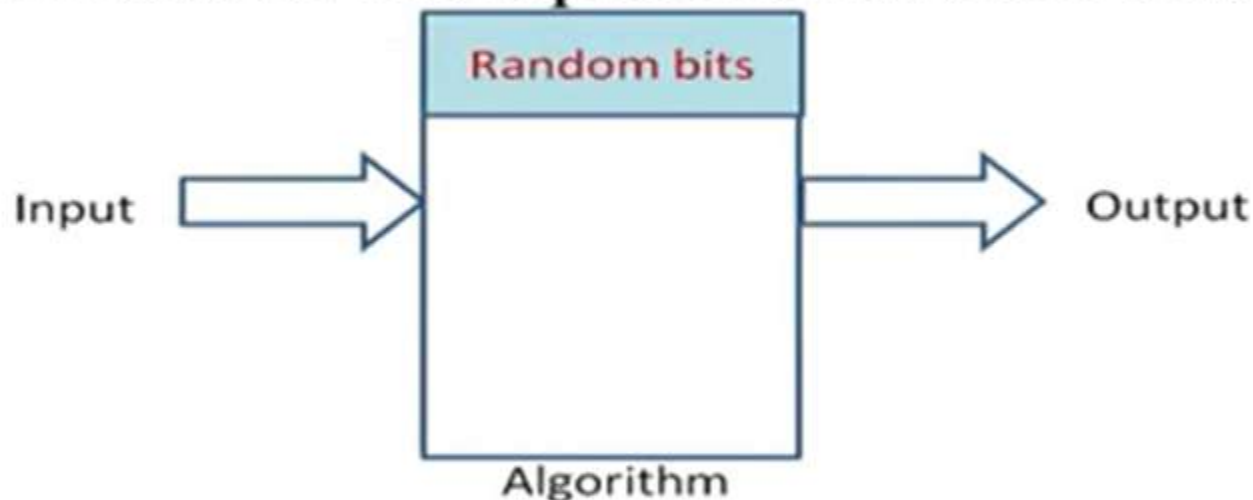


Randomized algorithms

MODULE-6

Randomized Algorithms

- **Randomized Algorithm:** The output or the running time are functions of the input and random bits chosen



- An algorithm that uses random numbers to decide what to do next anywhere in its logic is called Randomized Algorithm
- Typically, this randomness is used to reduce time complexity or space complexity in other standard algorithms

Randomized Algorithms

- **Advantage:**
 - For many problems, a randomized algorithm is the simplest and the fastest
 - Many NP-hard/NP Complete problems can be easily solvable

Randomized quick sort

- The quicksort technique is done by separating the list into two parts.
- Initially a pivot element is chosen by partitioning algorithm.
- The left part of the pivot holds the smaller values than pivot, and right part holds the larger value.
- After partitioning, each separate lists are partitioned using same procedure.
- In Randomized Quick Sort, we use a random number to pick the next pivot. Typically, this randomness is used to reduce time complexity or space complexity in other standard algorithms.

ALGORITHM

RANDOMIZED-PARTITION(A, p, r)

```

1   $i = \text{RANDOM}(p, r)$ 
2  exchange  $A[r]$  with  $A[i]$ 
3  return PARTITION( $A, p, r$ )

```

The new quicksort calls RANDOMIZED-PARTITION in place of PARTITION:

RANDOMIZED-QUICKSORT(A, p, r)

```

1  if  $p < r$ 
2       $q = \text{RANDOMIZED-PARTITION}(A, p, r)$ 
3      RANDOMIZED-QUICKSORT( $A, p, q - 1$ )
4      RANDOMIZED-QUICKSORT( $A, q + 1, r$ )

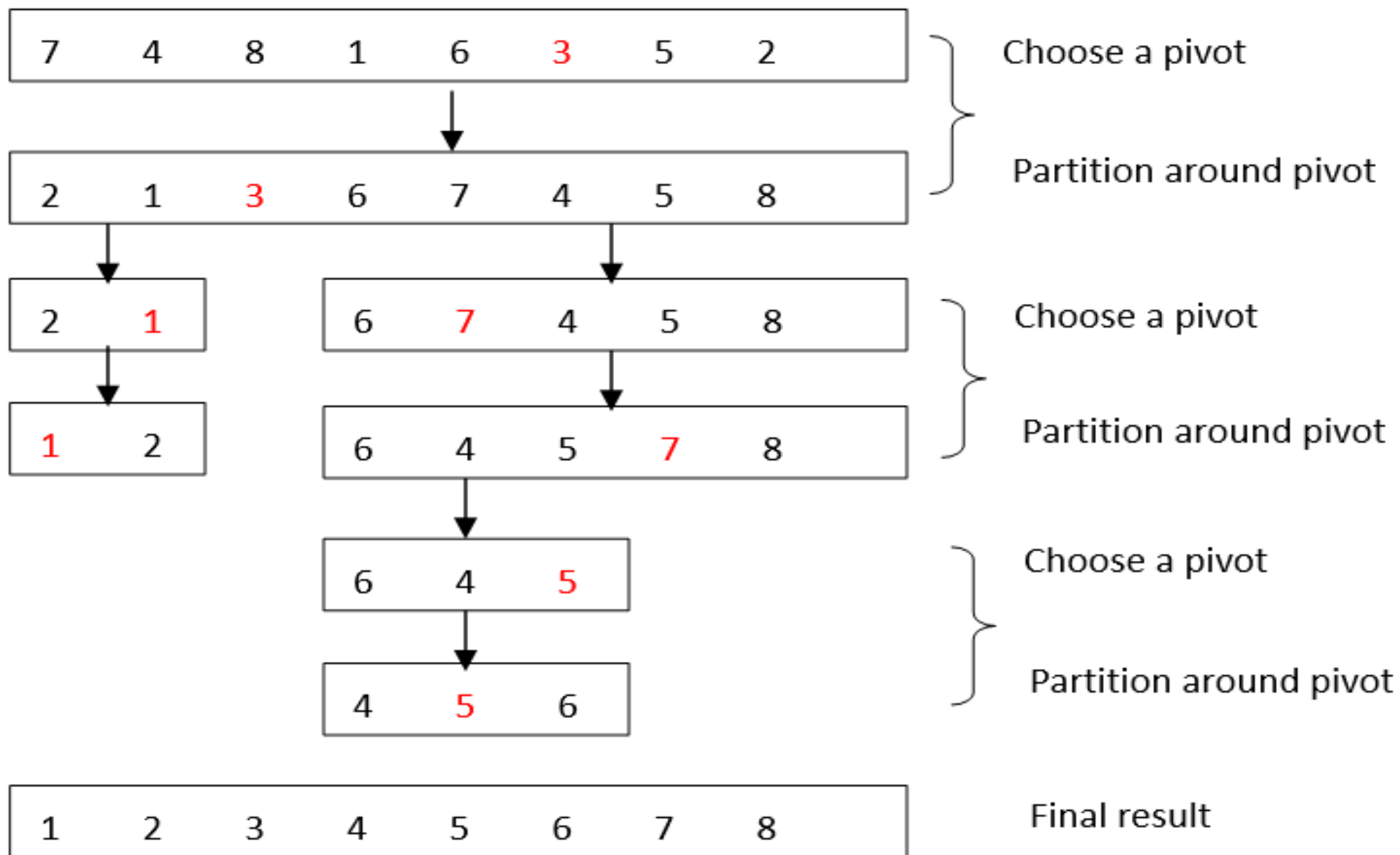
```

PARTITION(A, p, r)

```

1   $x = A[r]$ 
2   $i = p - 1$ 
3  for  $j = p$  to  $r - 1$ 
4      if  $A[j] \leq x$ 
5           $i = i + 1$ 
6          exchange  $A[i]$  with  $A[j]$ 
7  exchange  $A[i + 1]$  with  $A[r]$ 
8  return  $i + 1$ 

```



Complexity

The complexity is

Worst case time complexity: $O(n \log n)$

Space complexity: $O(n)$